

Unit 104 - Field Data Review

Mohawk Valley West Slope demonstration. Method **Tractor**, 111.5 ac, BAF 20 variable-radius cruise on a 300 ft grid, 59 plots, 611 trees tallied. Simulated data with planted recording errors; see docs/methods.md.

Table 1. Stand summary from the BAF 20 cruise, per acre

STAND DENSITY		STAND INFO	
Basal area	207 sq ft/ac (SE 2.0 %)	Type	Sierran mixed conifer
Trees per acre	165	CWHR size / density	4 / D (cover 85 %)
QMD	15.2 in	SDI	300 (52 % of max 572)
SAWTIMBER (>= 10 in)		BIOMASS (< 10 in)	
BA QMD	186 sq ft/ac 18.2 in	BA QMD	21 sq ft/ac 7.8 in
TPA	103	TPA	62
Volume	32.0 MBF/ac (5822 cu ft)	Volume	10.2 green tons/ac (341 cu ft)
TREATMENT TARGET		CRUISE DESIGN	
Leave	35 % of max SDI = ~138 sq ft/ac BA	Sampling error	3.9 % at 95 % (t = 2.00, n = 59)
Species (BA)	PP 66, WF 61, IC 32, DF 29, SP 20	Stratum standard	40 % - MEETS; 59 plots vs 20-plot practice: ok

Cubic volume by form factor 0.42 net of defect; Scribner board feet at 5.5 bf/cu ft (Keegan et al. 2010); green tons at 60 lb/cu ft. SDI in the summation form; maximum is the basal-area-weighted FVS Western Sierra species maximum. Sampling error at 95 % confidence with Student's t; stratum standard 40 % for tree-measurement sales (FSH 2409.12 ch. 40, 41.1).

Table 2. Stand table by 4-inch DBH class, per acre

DBH class, in	Trees	TPA	BA	cu ft	MBF
4-7.9	21	31.5	7.1	106	0.6
8-11.9	79	50.4	26.8	480	2.6
12-15.9	108	34.9	36.6	819	4.5
16-19.9	121	23.9	41.0	1133	6.2
20-23.9	81	10.5	27.5	860	4.7
24-27.9	78	7.4	26.4	917	5.0
28-31.9	38	2.7	12.9	511	2.8
32-35.9	30	1.6	10.2	440	2.4
36-39.9	24	1.1	8.1	370	2.0
40-43.9	12	0.4	4.1	197	1.1
44-47.9	7	0.2	2.4	124	0.7
48-51.9	6	0.2	2.0	104	0.6
52-55.9	2	0.0	0.7	35	0.2
56-59.9	2	0.0	0.7	34	0.2
60-63.9	2	0.0	0.7	33	0.2
All	611	164.8	207.1	6163	33.9

Table 3. Stock table by species, per acre

Species	Trees	TPA	BA	QMD	cu ft	MBF
PP	195	49.2	66.1	15.7	2026	11.1
WF	179	51.6	60.7	14.7	1815	10.0
IC	93	24.7	31.5	15.3	902	5.0
DF	86	20.6	29.2	16.1	871	4.8
SP	58	18.5	19.7	14.0	549	3.0
All	611	164.8	207.1	15.2	6163	33.9

Trees = trees tallied in the cruise; TPA, BA (sq ft/ac), cubic feet and Scribner MBF per acre by BAF expansion. Species: PP ponderosa pine, WF white fir, DF Douglas-fir, SP sugar pine, IC incense-cedar.

Unit 104: plots on the 300 ft grid (orange = QA flag)

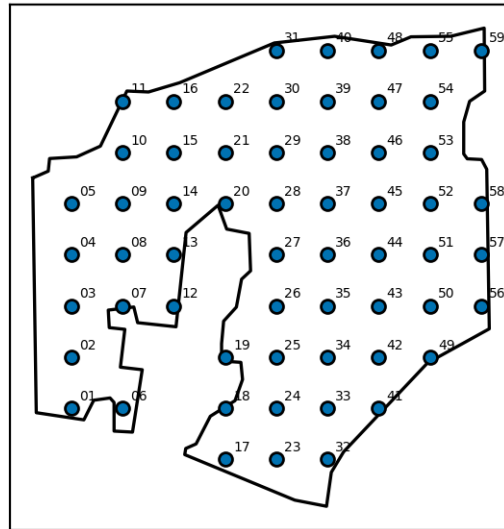


Figure 1. Plot locations with QA status.

QA findings: no issues found; unit accepted

Table 4. Plot list

Plot	Trees in	BA sq ft/ac	Slope %	Status
104-01	8	160	57	ok
104-02	12	240	36	ok
104-03	12	240	21	ok
104-04	11	220	35	ok
104-05	6	120	14	ok
104-06	10	200	100	ok
104-07	8	160	96	ok
104-08	11	220	34	ok
104-09	8	160	28	ok
104-10	9	180	33	ok
104-11	9	180	94	ok
104-12	10	200	105	ok
104-13	11	220	86	ok
104-14	10	200	21	ok
104-15	9	180	42	ok
104-16	8	160	37	ok
104-17	11	220	63	ok
104-18	12	240	69	ok
104-19	9	180	167	ok
104-20	12	240	148	ok
104-21	9	180	19	ok
104-22	11	220	30	ok
104-23	11	220	84	ok
104-24	10	200	83	ok
104-25	12	240	73	ok
104-26	10	200	134	ok
104-27	13	260	60	ok
104-28	11	220	25	ok
104-29	8	160	38	ok
104-30	11	220	17	ok
104-31	11	220	54	ok
104-32	11	220	11	ok
104-33	11	220	46	ok
104-34	8	160	50	ok
104-35	12	240	38	ok
104-36	13	260	34	ok
104-37	11	220	107	ok
104-38	11	220	60	ok
104-39	10	200	32	ok
104-40	11	220	36	ok
104-41	11	220	35	ok
104-42	12	240	37	ok
104-43	10	200	31	ok
104-44	12	240	10	ok
104-45	11	220	41	ok
104-46	9	180	123	ok
104-47	12	240	17	ok
104-48	9	180	55	ok
104-49	10	200	77	ok
104-50	6	120	40	ok
104-51	12	240	26	ok

Plot	Trees in	BA sq ft/ac	Slope %	Status
104-52	12	240	9	ok
104-53	11	220	94	ok
104-54	10	200	21	ok
104-55	12	240	13	ok
104-56	11	220	47	ok
104-57	11	220	28	ok
104-58	10	200	43	ok
104-59	9	180	15	ok